

CLASS XII · LECTURE 8

Visual Basic: Forms and Controls

Option II · Two periods · Text box, combo box, option button, check box

By the end of this lecture you can

- Name the parts of the VB design screen you use in every program.
- List the properties of a text box and of a combo box that the paper asks for.
- Choose a check box or an option button for the right kind of choice.
- Expand ADO, MDI, OLE, SQL and DBA.

The screen you design on

This option is Visual Basic 6, the form-based language in the BIEK paper, not VB.NET. You draw a **form**. On it you place **controls**. You set their **properties**. You write **code** in an event, usually the Click of a command button. The toolbox holds the controls. The properties window edits the selected control. The code window holds the procedures. A project is saved as a form file plus a project file. That is the environment. You do not need the menu history.

Control	What the user sees	Typical job
Label	Text the user cannot edit	A caption such as "Marks"
TextBox	A box the user can type in	Input, or a result you place in the Text property
CommandButton	A button	Runs the code in its Click event
OptionButton	A round choice	Exactly one from a group: male or female, or a payment method
CheckBox	A square yes/no	An independent choice. Several can be ticked. The value is checked or not, often thought of as 1 or 0
ListBox	A list that stays open	Pick one item from a visible list
ComboBox	A text box with a drop-down list	Pick an item, or type one, depending on Style

Read the question's verb. "One choice from a group" is an **option button**. "A yes or no that stands alone" is a **check box**. The paper sometimes phrases a check box as choosing yes (1) or no (0). That wording is the check box, because each box stores its own yes or no. It is not an option button, and it is not a command button.

Properties to memorise

Control	Properties worth writing
TextBox	Name, Text, Font, Enabled, Visible, MaxLength, MultiLine, PasswordChar, Locked
ComboBox	Name, Text, List, Style, Sorted, Enabled, Visible
CommandButton	Name, Caption, Enabled, Visible, Default, Cancel
Label	Name, Caption, Font, AutoSize
Form	Name, Caption, BackColor

Name is how the code talks to the control: `Text1.Text`. **Caption** is the words on a button or a label. A text box has **Text**, not **Caption**. **Enabled** = **False** shows the control but refuses clicks. **Visible** = **False** hides it. **PasswordChar** set to an asterisk hides a password as stars. **MaxLength** limits the characters. **MultiLine** = **True** allows more than one line. On a combo box, **List** is the set of items, **Text** is the current value, **Style** decides whether the user may type or only pick, and **Sorted** keeps the list in order.

ActiveX controls are extra controls you add to the toolbox, beyond the ones that are there at the start. You need the name and that sentence. You do not need to install one in the exam.

Full forms that belong to this option

Short form	Full form	One fact
ADO	ActiveX Data Objects	A library VB uses to open a database and read records
MDI	Multiple Document Interface	A parent window that holds several child windows
OLE	Object Linking and Embedding	Placing an object from another program, such as a sheet or a picture, into yours
SQL	Structured Query Language	The language of relational database questions
DBA	Database Administrator	The person responsible for the database

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. The default Step in For...Next is:

- A) 0
- B) 2
- C) 1
- D) -1

2. A control used to pick exactly one option from a group is the:

- A) Check box, when several may be on
- B) Option button
- C) Label
- D) Timer

Short answers

S1. List four properties of a text box.

S2. What is a combo box? List three properties.

S3. Write the full form of ADO, MDI and SQL.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. C. If you omit Step, the loop counts upward by 1. The loop itself is in the next lecture; the fact is a one-mark answer.

MCQ 2. B. Option buttons in one group are mutually exclusive.

S1. Name (the name in code), Text (the contents), MaxLength (the maximum number of characters), and PasswordChar (the character shown instead of the real text). Enabled, Visible, MultiLine and Font are equally acceptable.

S2. A combo box is a control that combines a text box with a drop-down list, so the user can pick an item or, depending on Style, type one. Three properties are Text, List and Style. Sorted is a fourth.

S3. ADO is ActiveX Data Objects. MDI is Multiple Document Interface. SQL is Structured Query Language. OLE is Object Linking and Embedding and DBA is Database Administrator, if the question offers those instead.